

THOMAS TO

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PROFESSIONAL SUMMARY

Full Stack AI Engineer with 7+ years building end-to-end systems across software engineering and applied machine learning. Translated a predictive machine learning revenue model into natural-language insight for non-technical finance and operations leadership, cutting decision cycles from 3+ hours to under 10 minutes and saving 500+ hours annually. Collaborates with technical and non-technical stakeholders to evaluate engineering trade-offs and deliver measurable business outcomes while mentoring engineers in statistical modeling and machine learning applied software engineering. Featured by International Business Times as a thought leader on AI-enabled drug discovery.

TECHNICAL SKILLS

- **Languages:** Python, C++, SQL, JavaScript, TypeScript, Google Apps Script, R, HTML, CSS, MATLAB
- **Web and APIs:** React, Next.js, Vue.js, Node.js, RESTful APIs, FastAPI, Tailwind CSS, Streamlit, Mermaid.js
- **AI and ML:** Generative AI, Large Language Models (OpenAI GPT, Anthropic Claude, Google Gemini, Snowflake Cortex), Retrieval-Augmented Generation, Prompt Engineering, Fine-Tuning, LangChain, LangGraph, LangSmith, MCP, n8n, Vector Databases, Embedding Models, Hugging Face, Model Evaluation, Guardrails, PyTorch, NumPy, pandas, OpenCV, TensorBoard, RDKit, Molecular Dynamics & Docking, Chemistry & Cheminformatics, pyRosetta, PyMOL, FoldIT, Rosetta, OpenFold3, Rosetta Foundry
- **Cloud and DevOps:** GCP, AWS, Docker, CI/CD, MLOps, Containerization, Kubernetes, Vercel, GitHub Actions, Model Monitoring, Infrastructure as Code, Observability, OpenXLA, JAX, CuPy
- **Data Management:** Snowflake, MongoDB, PostgreSQL, dbt, Fivetran, SAP S/4HANA, ETL/ELT Pipelines, Data Warehousing, Data Modeling, Tableau, Sigma

PROFESSIONAL EXPERIENCE

Founding AI Engineer | Open Source | Oakland, CA | Dec 2017 - Present

- Diagnosed a hardware lock-in bug blocking TPU and Apple Silicon contributors in an external open-source drug-discovery repository, then migrated its pipeline to the unified torch.accelerator API and shipped deterministic cross-backend seeding, earning a second unprompted contribution merged by the maintainer ([PR #24](#), [PR #25](#)).
- Architected a production Retrieval-Augmented Generation agent with LLM output validation and guardrails on Next.js and React using typescript to ship 0-to-1 from system design through open-source release and gaining adoption from 10+ external engineers with no reporting line into the project.
- Scaled independent technical thought leadership by automating a content pipeline (LLM orchestration, prompt engineering, output guardrails) through GitHub Actions CI/CD, sustaining 7 weekly publications while eliminating 5+ hours per week of manual production work.
- Extended technical range beyond an existing specialization to contribute distributed-training internals to OpenXLA, JAX, and Kubernetes, then applied the hardware-invariant mental model to ship deterministic reproducibility across CUDA, TPU, and Apple Silicon backends.
- Delivered a free AI research newsletter translating frontier machine learning research into practical takeaways for engineers and academics, by deploying a 4-bit quantized model on Hugging Face Infrastructure and an automated fallback for volume spikes, eliminating recurring token costs entirely.

Founding Fullstack Engineer | Canvanta Life Sciences | Emeryville, CA | Jan 2023 - Present

- Leads infrastructure and analytics strategy for Canvanta as its founding engineer, translating mathematical models and compliance requirements into decisions technical and executive stakeholders to act on by communicating multi-year cross-company technical strategy with parent organization STEMCELL Technologies.
- Translated a predictive machine learning revenue model into natural-language query access for non-technical finance and operations stakeholders by pairing it with a Retrieval-Augmented Generation agent on Snowflake, replacing a manual spreadsheet-limited process and cutting decision cycles from 3+ hours to under 10 minutes while saving 500+ hours annually.
- Aligned Canvanta and its 2,000-employee parent company STEMCELL Technologies around a shared GCP infrastructure rollout timeline with no reporting line into either organization, translating legal and compliance requirements into actionable terms for executives at both companies, rather than finalizing a plan and asking for ratification after the fact.
- Resolved a document digitization accuracy gap on 5,000+ handwritten laboratory records by fine-tuning Snowflake Arctic-TILT against systematically held-out evaluation data, reaching 95%+ accuracy and saving 1,000+ hours of manual transcription.
- Diagnosed a shared underlying need behind finance, operations, and lab teams' fragmented data requests, then consolidated them into one self-serve system spanning Python, dbt, and Snowflake with Streamlit and Tableau layers, cutting ad-hoc requests 70% and recovering 30+ production hours per week.
- Prevented a \$2M inventory stockout and \$200K in material waste by building data-driven demand forecasting models ahead of the risk becoming visible in existing reporting.

Software Engineer | Genentech | South San Francisco, CA | *Jun 2022 - Jan 2023*

- Translated scientific teams' workflow bottlenecks into full-stack internal tools built with Python REST APIs, React, Vue.js, and Node.js, streamlining processes for 5+ scientific teams and saving 15+ hours per week.
- Accelerated feature delivery from 4+ weeks to under 1 week by folding stakeholder feedback directly into the development cycle instead of collecting it at the end, across cross-functional teams.

Process Engineer | Genentech | Vacaville, CA | *Jun 2021 - Jun 2022*

- Operationalized manufacturing data infrastructure end-to-end, converting manual scientific workflows into automated systems adopted across production teams with Python, R, and SQL pipelines built for parallel execution and batch optimization, cutting processing time 99% from weeks to minutes.
- Digitized manual workflows into structured databases and reporting dashboards for 5+ cross-functional teams, saving 10+ hours per week and giving leadership real-time visibility into production metrics.

Research Engineer | UC Davis (Nandi, McDonald, Wan, Siegel Labs) | Davis, CA | *Sep 2019 - Jun 2021*

- Led computational method development spanning process optimization, image analysis, and protein engineering across four collaborating faculty labs using Python (NumPy, pandas) and parallel numerical algorithms to project \$63.2M in manufacturing cost reductions.
- Automated microscopy image analysis with a Python and OpenCV computer vision pipeline, processing 10,000+ images in hours versus weeks of manual counting.
- [Published](#) 3 novel protein variants for biomanufacturing by designing computational protein models with pyRosetta and PyMOL, validating designs through wet-lab testing and iterative optimization.

Teaching Assistant | University of California, Davis | Davis, CA | *Sep 2019 - Jun 2022*

- Modernized course technical infrastructure while extending instruction beyond assigned curriculum, translating complex programming and data-modeling concepts into accessible coding exercises and workshops for students migrating legacy MATLAB codebase to Python3 while developing supplemental coding exercises and debugging workshops covering data preprocessing, feature engineering, and iterative model development.

Laboratory Coordinator | Laney College | Oakland, CA | *Aug 2017 - Jun 2019*

- Built a hands-on computational curriculum for community college STEM students, translating foundational programming and data-analysis concepts into accessible coursework reaching 50+ students per semester through hands-on exercises in visualization and computational modeling.

PROJECTS

Real-Time WebSocket Puzzle Agent | *Jun 2026*

- Engineered an autonomous Node.js WebSocket agent with regex-first dispatch resolving 6 checkpoint types in sub-millisecond time against a 4-second server deadline, integrating a custom recursive-descent math parser, Wikipedia REST API, and Google Gemini 2.5 Flash function-calling fallback, validated by 107 test cases.

Embedded Edge AI on Raspberry Pi

- Deployed NVIDIA Alphamayo VLM+A models on Raspberry Pi for real-time edge inference at sub-200ms latency with zero cloud dependency, applying model quantization and Docker containerization.

Agentic Video Editing Pipeline

- Shipped a 0-to-1 open-source screen recording and editing platform reducing video production time by 80%, architecting an agentic pipeline with FFmpeg for automated video processing, Google Chirp 3 for speech-to-text transcription, and YouTube OAuth API for SEO-optimized publishing.

Document (ETL) Form Chrome Extension | *Jan 2026*

- Built a Chrome extension as a service paired with a Vercel-deployed React/Next.js frontend, implementing hashing, encryption, and Chrome security permissions for targeted HTML parsing across sheet URLs, in TypeScript with Google ADK and Anthropic Claude Code. Eliminating manual form filling for law firm documents and passport records by shipping an agentic ETL pipeline processing uploads through in-memory storage and auto-populating target ingestion forms, reducing a multi-hour manual process to seconds.

MEDIA & PUBLICATIONS

"Thomas To on How AI-Enabled Biology is Advancing Drug Discovery and Scientific Innovation" | International Business Times | *Jun 2026*

- Featured as a thought leader on AI in scientific research, discussing AI as a support tool for scientific decision-making rather than a replacement for empirical biology and scientific expertise. Made the case for specialized systems trained on scientific data over general conversational AI.

"Design to Data for mutants of beta-glucosidase B from Paenibacillus polymyxa: L171G, L171V and L171W" | McNair Scholars Program, UC Davis Genome Center | Mentored by Dr. Justin Siegel and Ashley Vater

EDUCATION

Bachelor of Science, Biochemical Engineering | University of California, Davis | Davis, CA | *Sep 2019 - Jun 2022*